



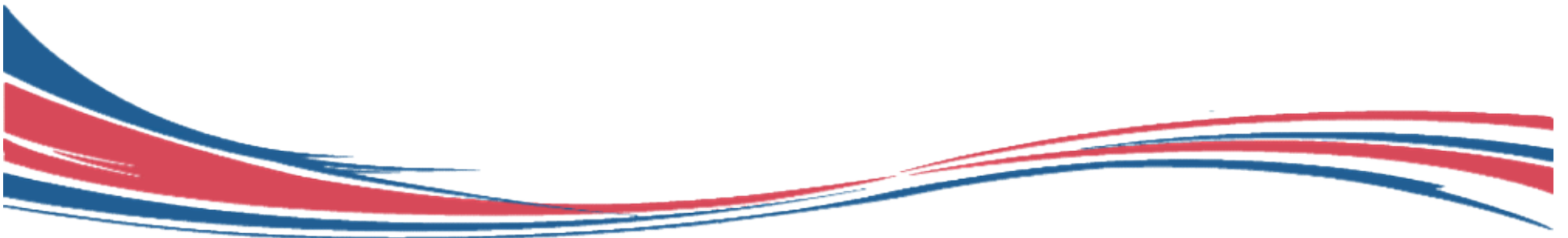
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PGA

Peak Ground Acceleration





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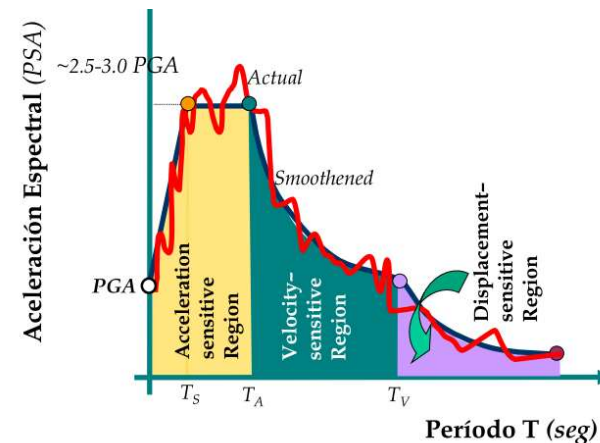
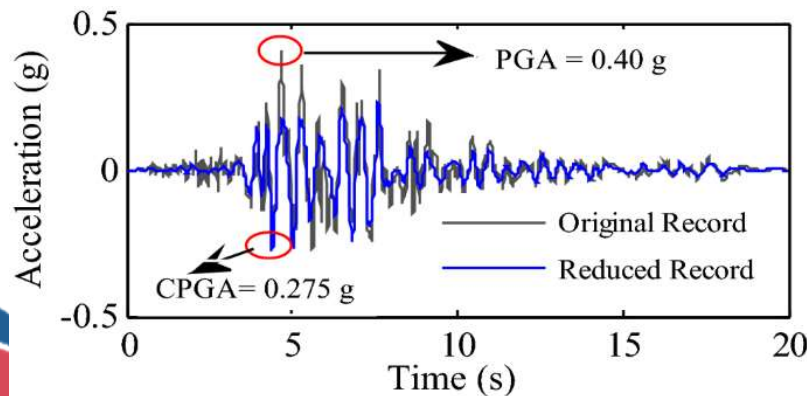


PGA:

La aceleración máxima del suelo (PGA), también denominada Aceleración pico del terreno es igual a la aceleración máxima del suelo que se produjo durante un terremoto en un lugar determinado.

La PGA es igual a la amplitud de la aceleración absoluta más grande registrada en un acelerograma en un sitio durante un terremoto en particular.

La PGA es la ordenada del espectro de aceleraciones, es un parámetro importante en ingeniería sísmica donde también es conocido como una medida de intensidad. Se expresa en cm/s^2 ó % g.

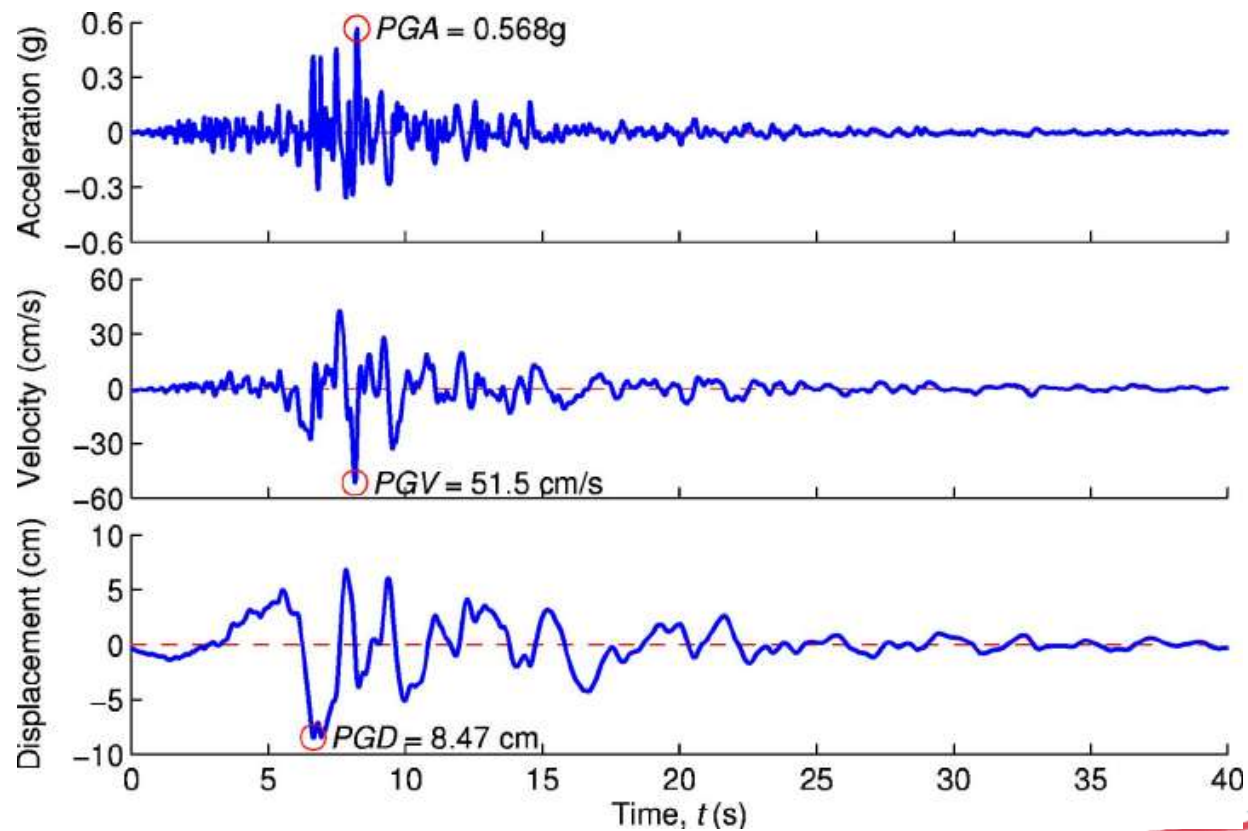




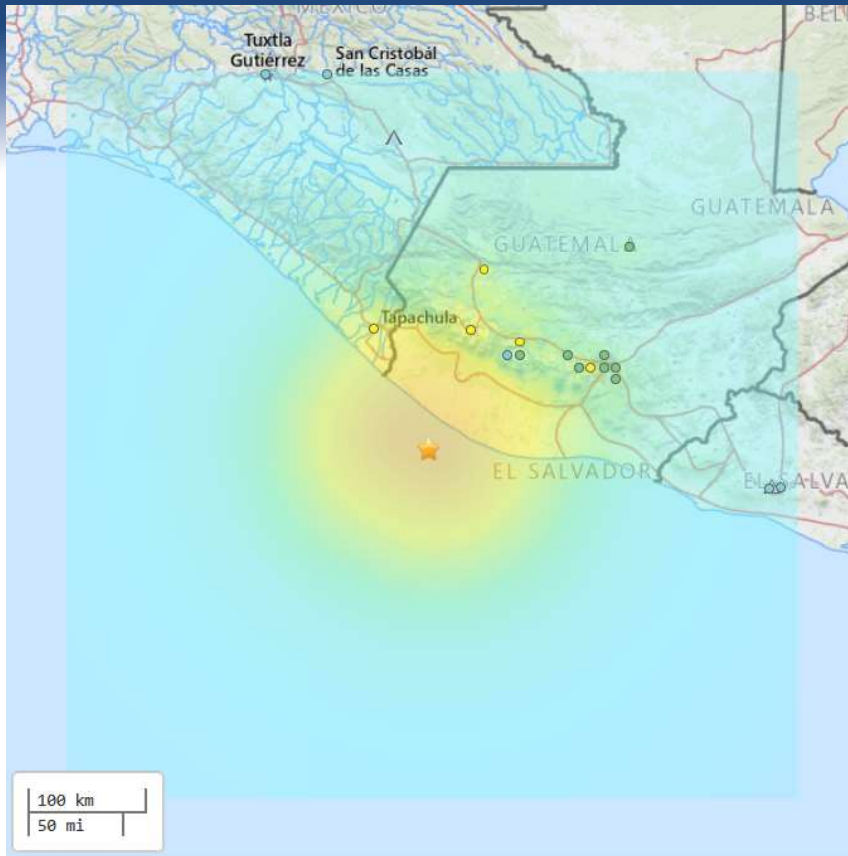
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Sismo 7 de noviembre de 2012



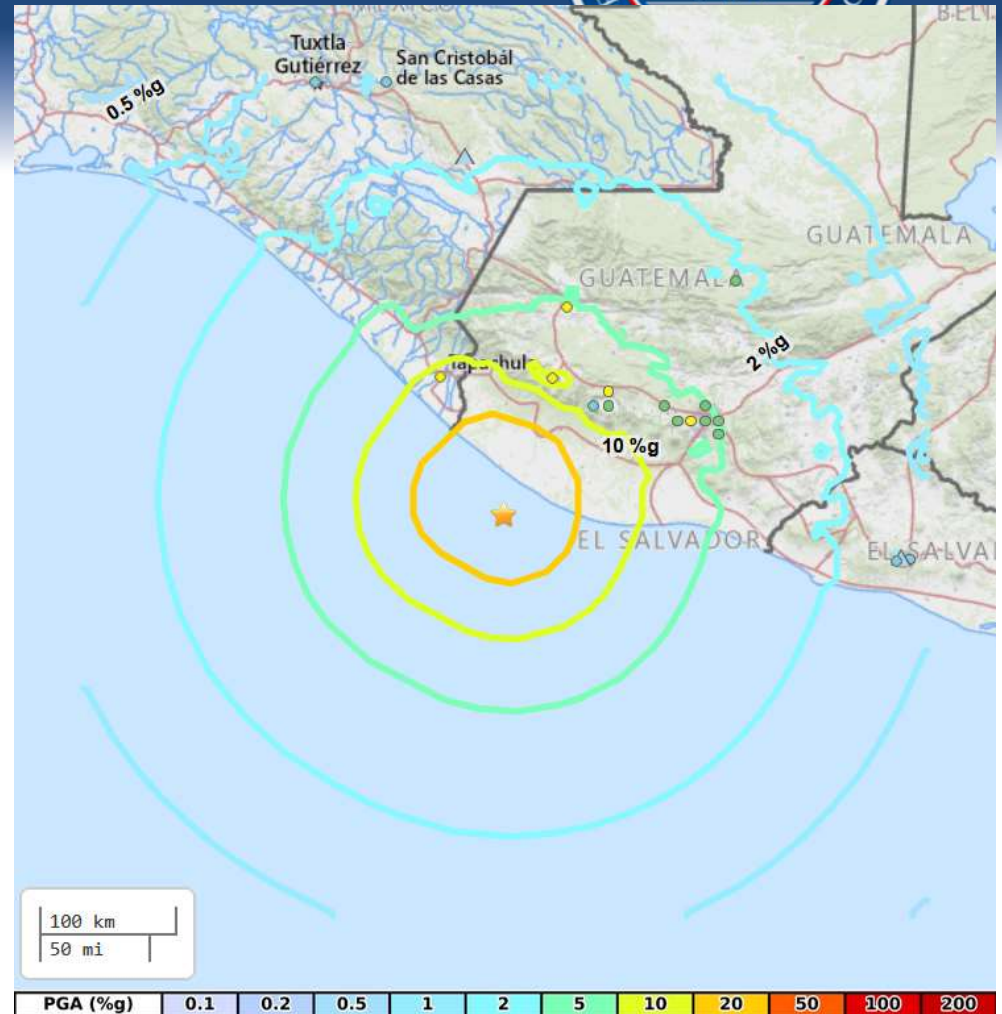
SHAKING	Not felt	Weak	Light	Moderate	Strong	Very strong	Severe	Violent	Extreme
DAMAGE	None	None	None	Very light	Light	Moderate	Moderate/heavy	Heavy	Very heavy
PGA(%g)	<0.0464	0.297	2.76	6.2	11.5	21.5	40.1	74.7	>139
PGV(cm/s)	<0.0215	0.135	1.41	4.65	9.64	20	41.4	85.8	>178
INTENSITY	I	II-III	IV	V	VI	VII	VIII	IX	X+

Scale based on Worden et al. (2012)

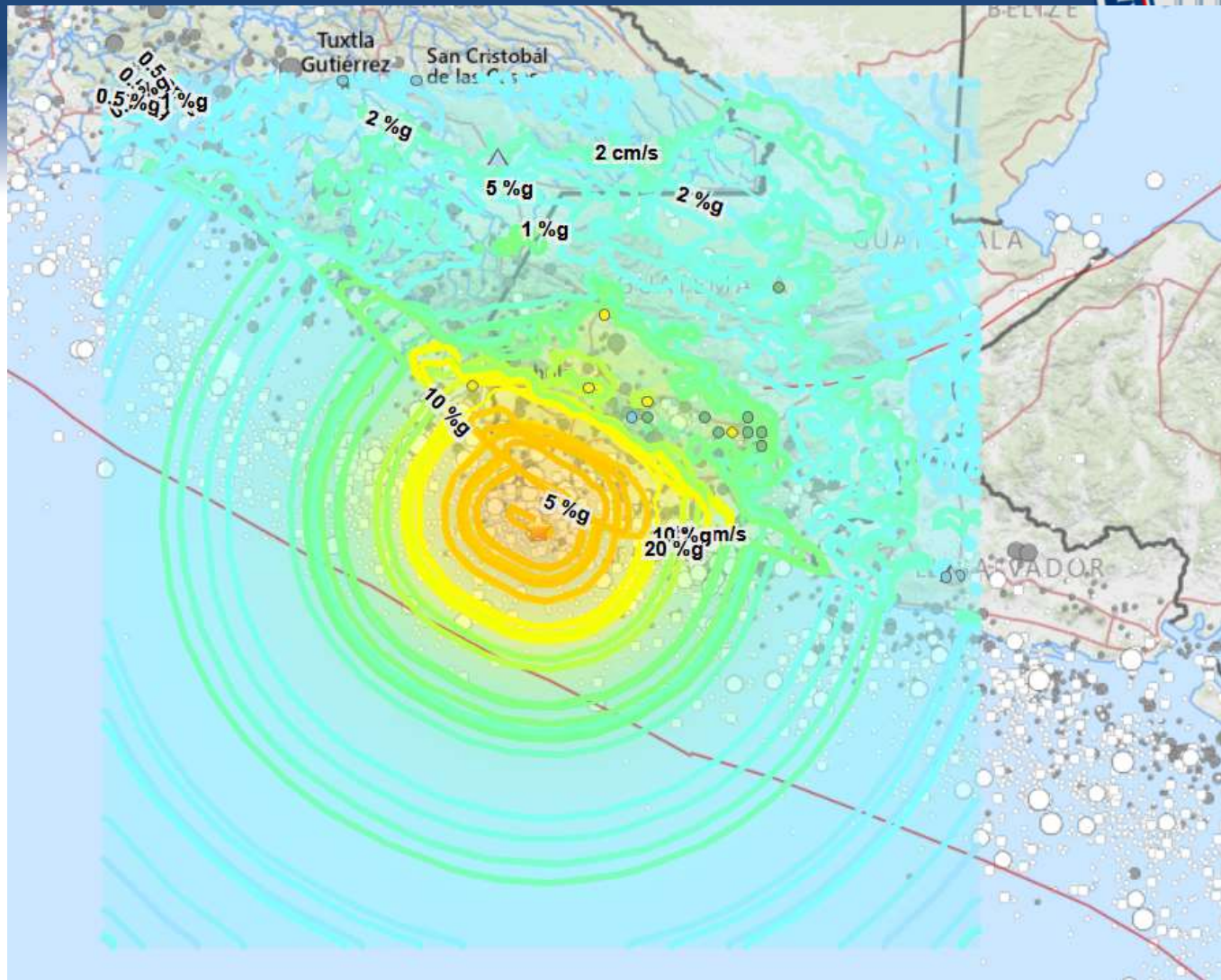
Version 1: Processed 2020-10-19T17:50:42Z

△ Seismic Instrument ○ Reported Intensity

★ Epicenter □ Rupture

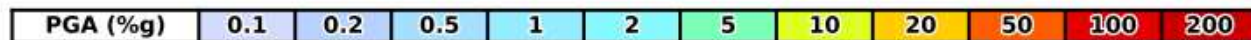
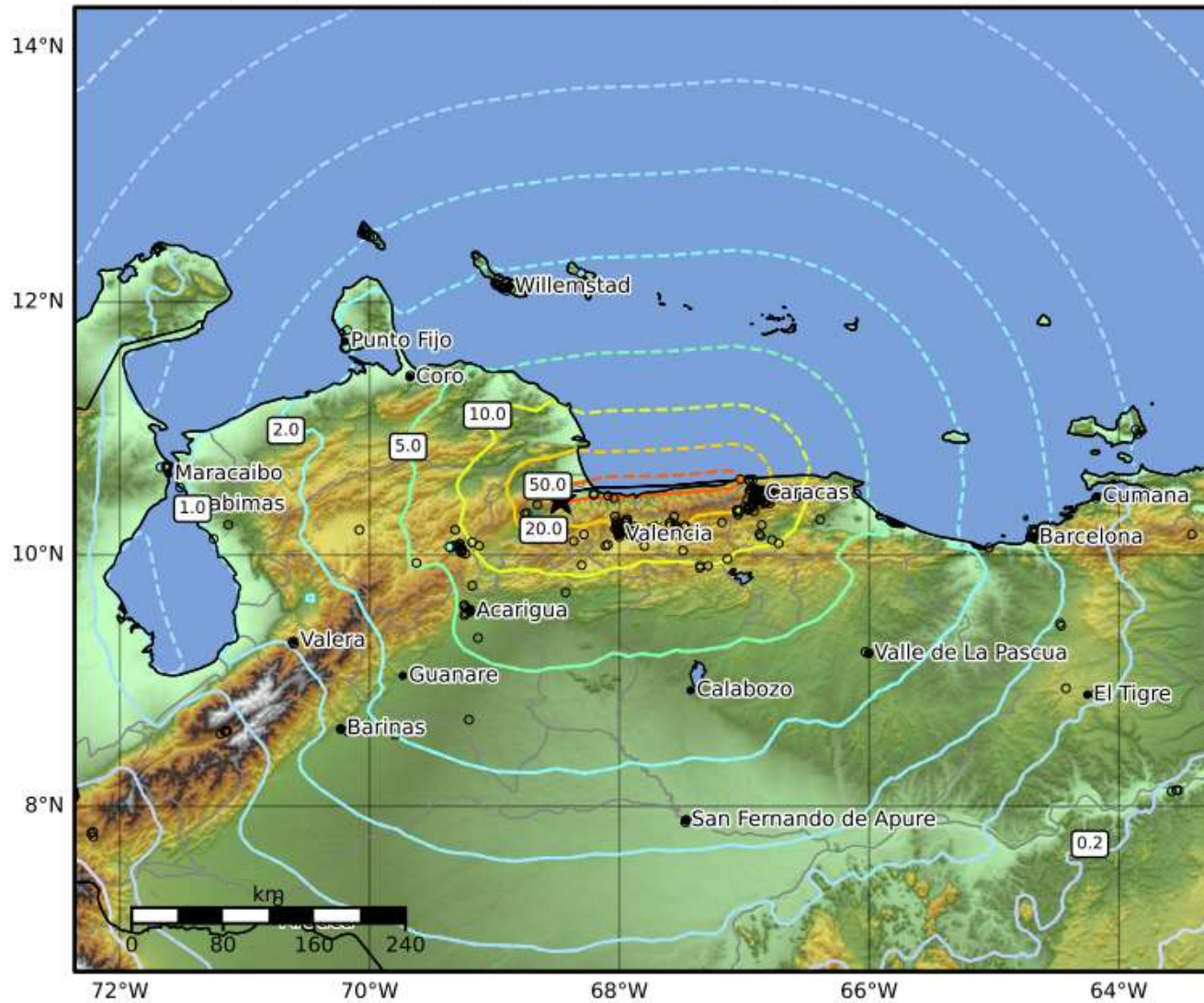


PGA (%g) 0.1 0.2 0.5 1 2 5 10 20 50 100 200



Curvas de atenuación de PGA

Peak Ground Acceleration Map USGS
 ShakeMap: 28 km SE of Yumare, Venezuela
 Jun 24, 2026 22:05:11 UTC M7.5 N10.44 W68.47 Depth: 10.0km ID:us6000t7zp



Scale based on Worden et al. (2012)

△ Seismic Instrument ○ Reported Intensity

★ Epicenter — Rupture

Version 8: Processed 2026-06-26T20:13:57Z



Venezuela
 24-junio-2026

Curvas de
 atenuación de
 PGA



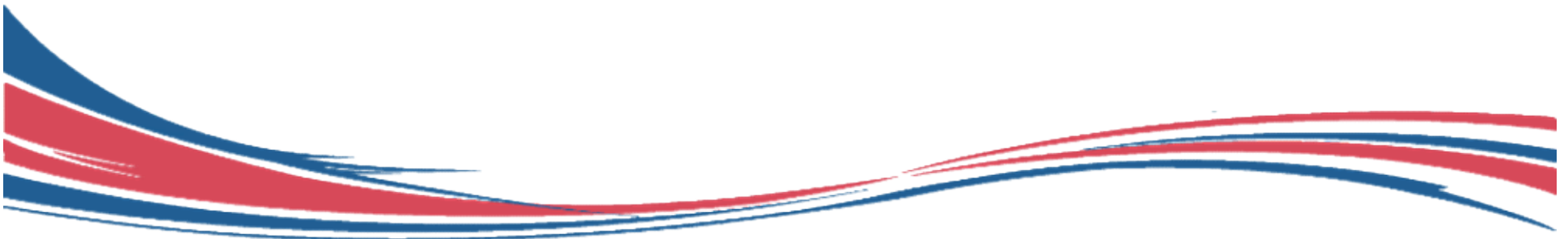
Calculo de PGA

ECUACIONES PREDICTIVAS

Formula de Joyner y Boore (1993)

$$\log PGA = -1.229 + 0.227M - \log^* \sqrt{R^2 + 44.225} - 0.0023 * \sqrt{R^2 + 44.225}$$

PGA en %g

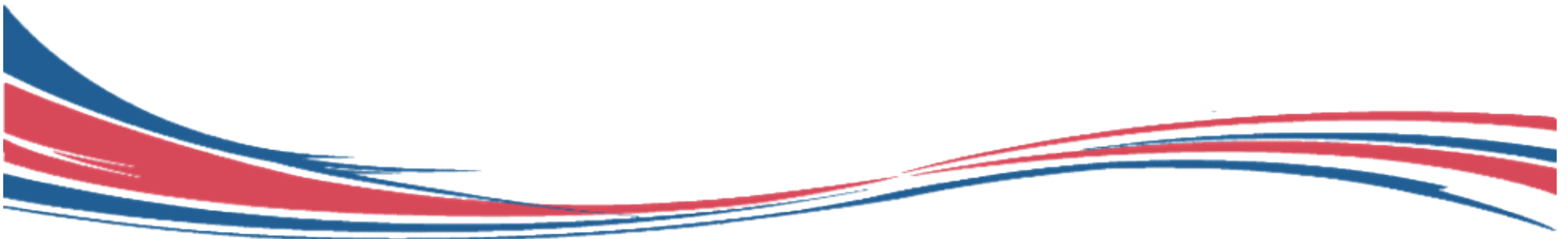




Formula de Cliement et al., (1994)

$$\ln PGA = -1.687 + 0.553M - 0.537 \ln R - 0.00302R$$

PGA en m/s^2

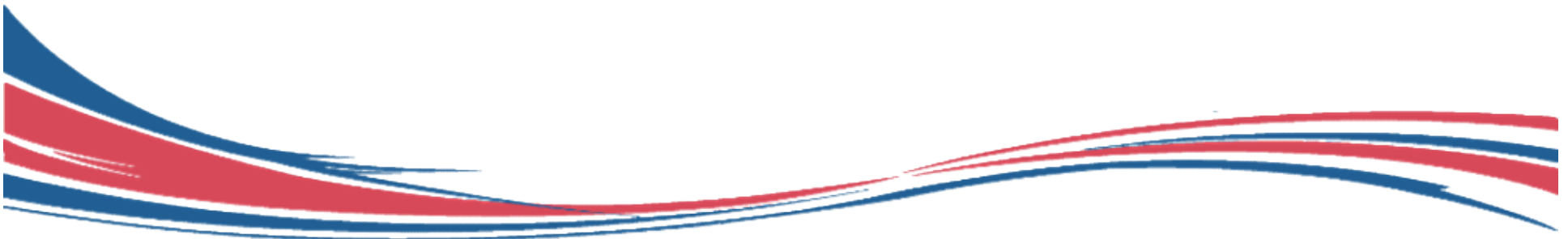




Cornell et.al. (1979)

$$\ln PGA = 6.74 + 0.859M - 1.8 * \ln(R + 25)$$

PGA en cm/s^2



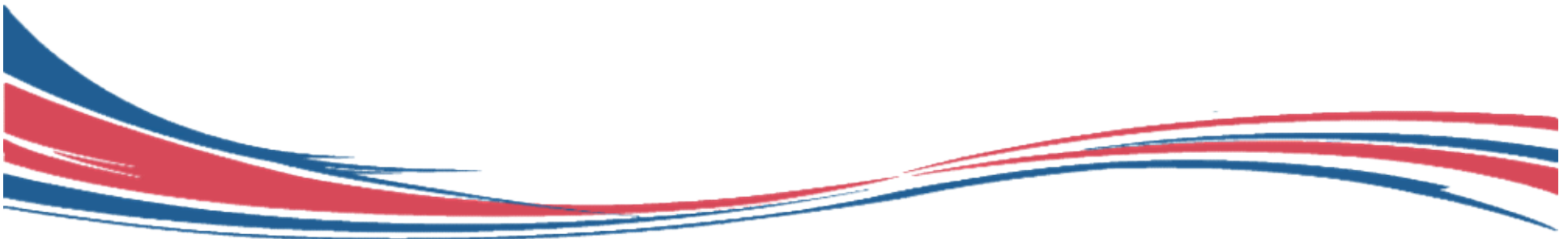


%Campbell (1981)

$$\ln PGA = -4.141 + 0.868M - 1.09 * \ln(R + 0.0606e^{0.7M})$$

PGA en %g

$R \leq 50$ km; $M = (5.0 \text{ A } 7.7)$



Ejemplo

$$|R| = \left| \frac{60 - 7(15) + 280}{- \sqrt{50}} \right| = 33.23 \text{ km}$$

Calculo de la magnitud momento (Mw)

$$\text{Normal} \quad 15 \quad M_w = 4.86 + 1.32 \log L$$

$$M_w = 4.86 + 1.32 \log_{10}(70.71) = 7.30$$

$$M_w = 7.30$$

Calcule la PGA, considerando los siguientes datos:

Falla A:

Mw=7.301 → Magnitud momento

R=33.234 Km → Distancia del epicentro a la falla

$$\log PGA = -1.229 + 0.227 * 7.301 - \log^* \sqrt{33.234^2 + 44.225} - 0.0023 * \sqrt{33.434^2 + 44.225}$$

$$\log PGA = -1.1801$$

$$PGA = 0.066g$$

$$PGA \text{ (gals)} = 64.746$$

Formula de Joyner y Boore (1993)

$$\log PGA = -1.229 + 0.227M - \log^* \sqrt{R^2 + 44.225} - 0.0023 * \sqrt{R^2 + 44.225}$$

PGA en %g



Falla A

$M=7.301$

$R=33.234 \text{ Km}$

$$\ln PGA = -1.687 + 0.553 * 7.301 - 0.537 \ln 33.234 - 0.00302 * 33.234$$

$$\ln PGA = 0.3687$$

$$PGA = 1.4458 \frac{m}{s^2} = \frac{1.4458}{9.81} = 0.1474g$$

Formula de Cliement et al., (1994)

$$\ln PGA = -1.687 + 0.553M - 0.537 \ln R - 0.00302R$$

PGA en m/s^2



Falla A

$M=7.301$

$R=33.234 \text{ Km}$

$$\ln PGA = 6.74 + 0.859 * 7.301 - 1.8 * \ln(33.234 + 25)$$

$$\ln PGA = 5.6955$$

$$PGA = 297.525 \frac{\text{cm}}{\text{s}^2} = \frac{297.525}{981} = 0.3033g$$

Cornell et.al. (1979)

$$\ln PGA = 6.74 + 0.859M - 1.8 * \ln(R + 25)$$

PGA en cm/s^2





Falla A

$M=7.301$

$R=33.234 \text{ Km}$

$$\ln PGA = -4.141 + 0.868 * 7.301 - 1.09 * \ln(33.234 + 0.0606e^{0.7*7.301})$$

$$\ln PGA = -1.9105$$

$$PGA = 0.148g$$

Campbell (1981)

$$\ln PGA = -4.141 + 0.868M - 1.09 * \ln(R + 0.0606e^{0.7M})$$

PGA en %g

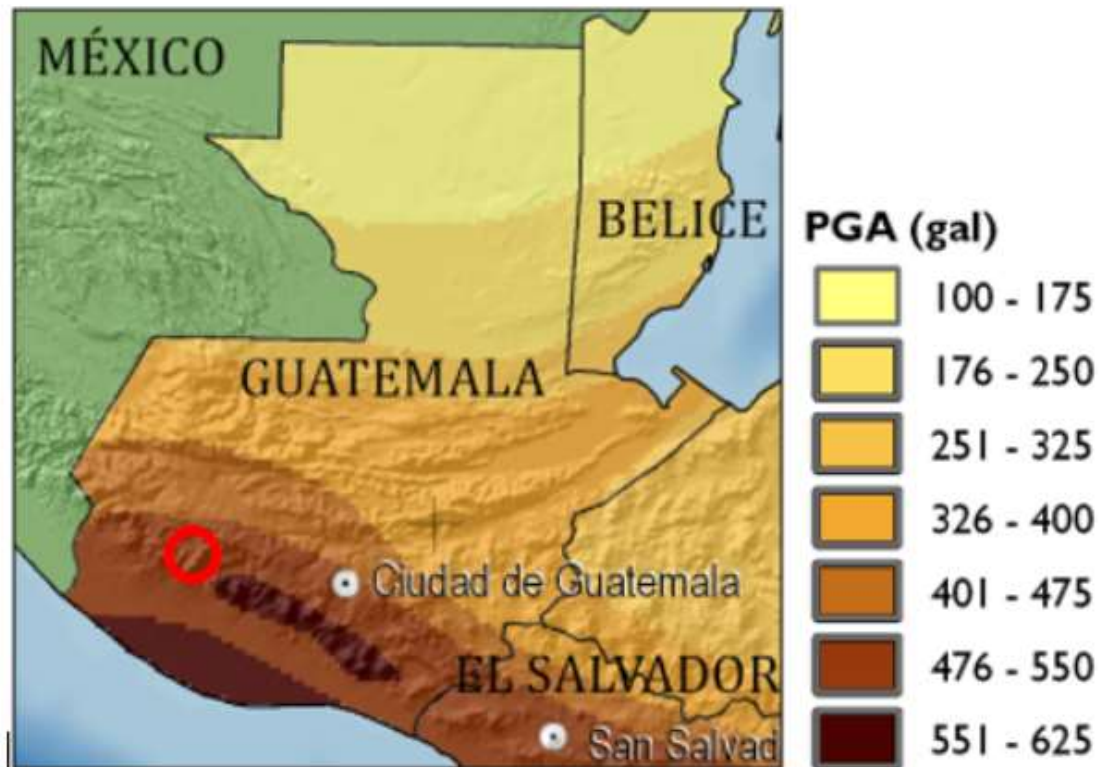


Falla A

M=7.301

R=33.234 Km

Formula	PGA
Joyner y Boore (1993)	0.066g
Cliement et.al. (1994)	0.1474g
Cornell et.al. (1979)	0.3033g
Campbell (1981)	0.148 g



Mapa de peligrosidad de Guatemala, en términos de PGA, en gales, para periodo de retorno de 475 años (Benito et al., 2010). El círculo rojo indica la ubicación de Quezaltenango

El gal (galileo) es la unidad de medida de aceleración, utilizada en geofísica. Su símbolo es Gal y equivale a 1 cm/s^2 .

$$g = 981 \text{ cm/s}^2$$

Peten	
min	max
0.10 g	0.17g

Quetgo	
min	max
0.48g	0.56g



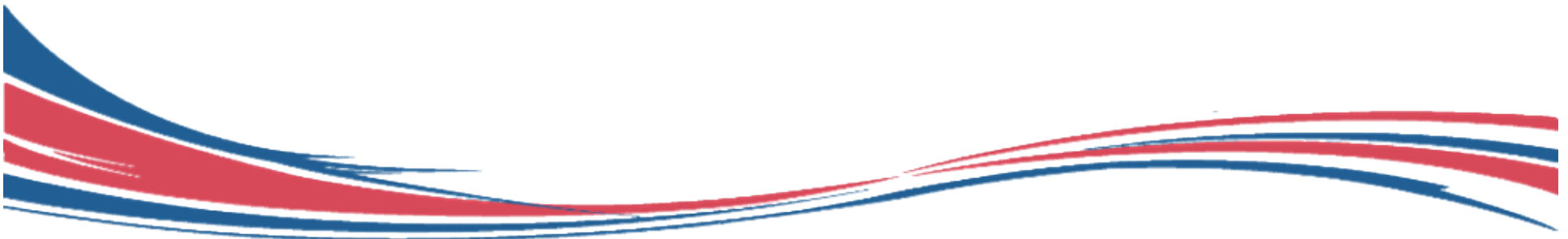
OTRAS ECUACIONES PREDICTIVAS

Toro et.al (1994) $R \leq 100$ km; $M_w > 8.0$

Subduction zone earthquakes generally occur at greater hypocentral depths than earthquakes that occur on transform faults. Consequently, the seismic waves that emanate from subduction zone earthquakes follow different paths from those of transform faults. Youngs et al. (1988) used strong-motion measurements obtained on rock from 60 earthquakes and numerical simulations of $M_w \geq 8$ earthquakes to develop a subduction zone attenuation relationship:

$$\ln \text{PHA}(g) = 19.16 + 1.045 M_w - 4.738 \ln [R + 205.5 \exp(0.0968 M_w)] + 0.54 Z_f \quad \sigma_{\ln \text{PHA}} = 1.55 - 0.125 M_w \quad (3.28)$$

where R is the closest distance to the zone of rupture in kilometers and Z_f is 0 for interface events and 1 for intraslab events.



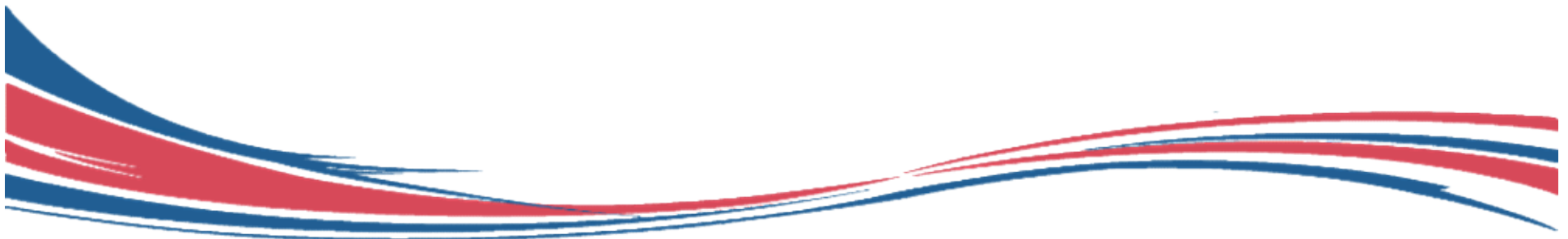


OTRAS ECUACIONES PREDICTIVAS

Campbell & Bozrgnia (1994) $R \leq 60$ km; $M = (4.7 \text{ a } 8.1)$

$$\begin{aligned} \ln \text{PHA(gals)} = & -3.512 + 0.904 M_w - 1.328 \ln \sqrt{R^2 + [0.149 \exp(0.647 M_w)]^2} \\ & + (1.125 - 0.112 \ln R - 0.0957 M_w) F + (0.440 - 0.171 \ln R) S_{SR} \\ & + (0.405 - 0.222 \ln R) S_{HR} \end{aligned} \quad (3.25)$$

$$\sigma_{\ln \text{PHA}} = \begin{cases} 0.889 - 0.0691 M & M \leq 7.4 \\ 0.38 & M > 7.4 \end{cases}$$





OTRAS ECUACIONES PREDICTIVAS

Boore et.al (1993) $R \leq 100$ km; $M = (5.0 \text{ a } 7.7)$

$$\log \text{PHA}(g) = b_1 + b_2(M_w - 6) + b_3(M_w - 6)^2 + b_4R + b_5 \log R + b_6G_B + b_7G_C \quad (3.26)$$

where $R = \sqrt{d^2 + h^2}$, d is the closest distance to the surface projection of the fault in kilometers, and

$$G_B = \begin{cases} 0 & \text{for site class A} \\ 1 & \text{for site class B} \\ 0 & \text{for site class C} \end{cases} \quad G_C = \begin{cases} 0 & \text{for site class A} \\ 0 & \text{for site class B} \\ 1 & \text{for site class C} \end{cases}$$

Table 3-5 Definitions of Site Classes for Boore et al. (1993) Attenuation Relationship

Site Class	$\bar{v}_s$ in Upper 30 m (100 ft)
A	> 750 m/sec (2500 ft/sec)
B	360–750 m/sec (1200–2500 ft/sec)
C	180–360 m/sec (600–1200 ft/sec)

Table 3-6 Coefficients for Boore et al. (1993) Attenuation Relationship

	Component								
	b_1	b_2	b_3	b_4	b_5	b_6	b_7	h	$\sigma_{\log \text{PHA}}$
Random	-0.105	0.229	0.0	0.0	-0.778	0.162	0.251	5.57	0.230
Larger	-0.038	0.216	0.0	0.0	-0.777	0.158	0.254	5.48	0.205

Toro et.al (1994) $R \leq 100$ km; $M = (5.0 \text{ a } 7.7)$

$$\ln \text{PHA}(g) = 2.20 + 0.81(M_w - 6) - 1.27 \ln R_m + 0.11 \max\left(\ln \frac{R_m}{100}, 0\right) - 0.0021 R_m \quad \sigma_{\ln \text{PHA}} = \sqrt{\sigma_M^2 + \sigma_R^2} \quad (3.27)$$

where $R_m = \sqrt{R^2 + 9.3^2}$, R is the closest horizontal distance to the earthquake rupture (in km), $\sigma_M = 0.36 + 0.07(M_w - 6)$, and

$$\sigma_R = \begin{cases} 0.54 & \text{for } R < 5 \text{ km} \\ 0.54 - 0.0227(R - 5) & \text{for } 5 \text{ km} \leq R \leq 20 \text{ km} \\ 0.20 & \text{for } R > 20 \text{ km} \end{cases}$$

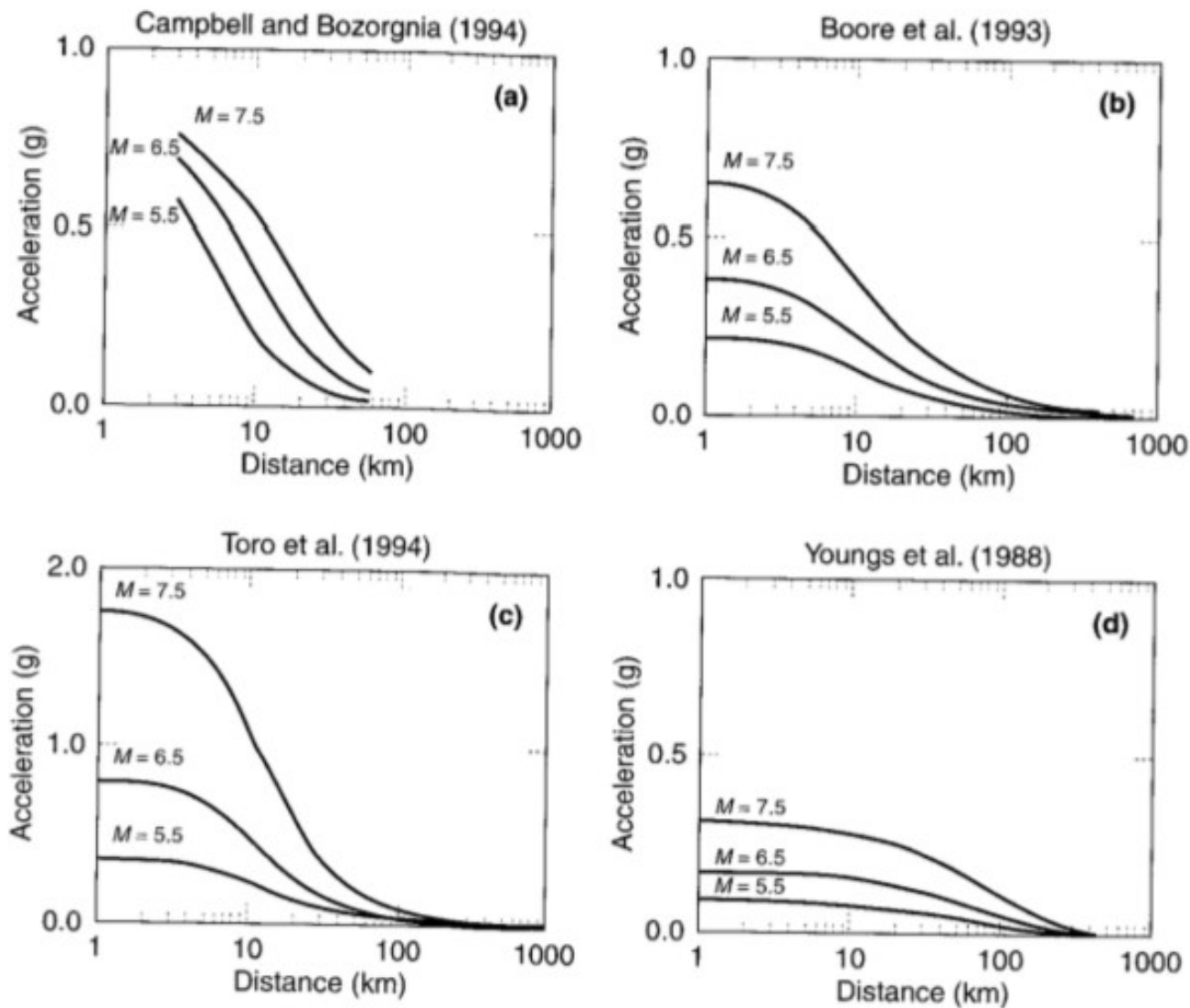


Figure 3.22 Variation of peak horizontal acceleration with distance for $M = 5.5$, $M = 6.5$, and $M = 7.5$ earthquakes according to various attenuation relationships: (a) Campbell and Bozorgnia (1994), soft rock sites and strike-slip faulting; (b) Boore et al. (1993), site class B; (c) Toro et al. (1994); and (d) Youngs et al. (1988), intraslab event.